CSE 342 | class - 01
Lab 01 | 4th March 2026

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1 set question
(built-in → 1,
custom → 1)

Exception Handling

Prevents a program to terminate abnormally.
Notifies the programmer about the error.

Java's built-in exception handling classes.
There are also custom exceptions

Exception Classes
- Built-in
- Custom

Exception Keywords:

1. Try → যেখানে exception creation এর chance আছে।
2. Catch → যে exception catch করবে (capture)
3. Throw → explicitly create an exception
4. Throws
5. Finally → executes regardless there's an exception

$\boxed{\text{throw object;}}$

↳ throw new ArithmeticException();

```
try {
    //...
}
catch (Exception e) {
    //...
    System.out.println(e);
}
```

```
try {...}
catch {...}
finally {...}
```

ToString()

Throwable
- error → Stack Overflow
- exception → RuntimeException

'Exception' is under 'lang' package
import java.lang;

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RuntimeException has many exceptions
↳ these are unchecked

RuntimeException

ArithmeticException

Divide by Zero

NullPointerException

String s = null;
×
↳ s.length()
↳ exception

ArrayIndexOutOfBounds Exception

```
int a[]={1,2,3};
a[10]=100;
```

```
public class Main{
  public static void main(String[] args){
    int a=10, b=0, c=0;
    c=a/b;
  try{
    ~~System.out.println(a+" "+b+" "+c);~~
  } catch (ArithmeticException e){ sop(e); }
finally{ System.out.println(a+" "+b+" "+c);
  }
```

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Multiple 'catch' can be used in a single try block, to handle multiple exception

```
String s=null;
try{
    sop(s.length());
}
catch(ArithmeticException e){
    sop(e);
}
catch(NullPointerException e){
    sop(e);
}
```

UNION exception → JDK 7+

```
catch(ArithmeticException | NullPointerException e){...}
```

NESTED Try

```
try {
    sop(s.length());
    try { c=a/b;
    } catch (ArithmeticException e){
        sop(e);
    }
} catch (NullPointerException e){
    sop(e); }
```

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```
public class exception {
    static void check (int a){
        if (a <0){
            throw new ArithmeticException("Neg number");
        }else{
            System.out.println(a*a);
        }
    }

    public static void main(String[] args){
        try{
            check(10);
            check(-2);
        } catch (ArithmeticException e){
            System.out.println(e.toString);
        }
    }
}
```

this method can throw exceptions

Why 'throws'?

```
static void check (int a) throws ArithmeticException,... {
    //...
}
```

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```
class exception extends Exception {
    exception(String s){
        super(s); // built-in constructor
    }
}

class abc{
    int age ;
    abc(int age){
        this.age = age;
    }
    void check() throws exception {
        If(age<18)
            throw new exception("No eligible");
        else
            System.out.println ("Eligible");
    }
}
public class Main{
    public static void main(String[] args){
        abc ob1 = new abc (12);
        try {
            ob1.check();
        } catch(exception e){
            ~~Serial~~ System.out.println ("—"+e); // e.getMessage() ;
        }
    }
}
```

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```
class exp2 extends Exception{
    exp2(String s){
        super(s);
    }
}

class Grading{
    int marks;
    Grading(int m){
        marks = m;
    }
    void check() throws exp2{
        if(marks < 0 || marks > 100)
            throw new exp2("Marks out of bound");
        else
            System.out.println("Marks within the range");
    }
}

public class Main{
    public static void main(String[] args){
        Grading g = new Grading(122);
        try{
            g.check();}
        catch(exp2 e){
            System.out.println(e.getMessage());
        }
    }
}
```

```
s CustomException extends Exception{
  CustomException(String s){
      super(s);
  }
}

class BankAccount{
  int amount;
  BankAccount(){
      amount =0;
  }
  Bank_Account(int a){
      amount = a;
  }
  void deposit(int a){
      amount += a;
  }
  void withdraw(int a) throws CustomException{
      if(amount < a || (amount-a) < 0){
          throw new CustomException("Insufficient Funds.");
      }else{
          amount -= a;
      }
  }
}

public class Main{
  public static void main(String[] args){
      BankAccount acc = new BankAccount(1000);
      acc.deposit(1000);
      try{ acc.withdraw(3000); }
      catch(CustomException e){System.out.println(e.getMessage());}
  }
}
```

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